Q-INTEL

QUICK-COMMERCE DELIVERY INTELLIGENCE

BUSINESS REPORT

Applied Machine Learning & Operations Analytics

An integrated decision-support prototype for delivery-time prediction, late-order risk detection, geographic demand analysis and courier performance segmentation.

BUSINESS CHALLENGE

Unpredictable delivery times, late-order exposure, uneven operational workload and uncertainty about store capacity.

PROPOSED RESPONSE

Combine supervised learning, unsupervised clustering, model evaluation and evidence-linked management actions.

REPORT PURPOSE

This report explains the operational problem, data and feature strategy, modelling approach, geographic and courier intelligence, and recommended management actions. It accompanies the runnable notebook, model comparison evidence and Streamlit demonstration.

PROJECT POSITIONING

Q-INTEL is a prototype decision-support system. It uses a public food-delivery dataset as a proxy for quick-commerce operations; restaurant locations act as store-location proxies. The data supports historical analysis and predictive experiments, but does not prove grocery dark-store profitability, live rider dispatch performance or realised cost savings.

Report contents

Page	Section	What it covers
2	Executive Summary	Business objectives, core analytical components, success criteria and intended users
3	Data, Reliability & Features	Data reliability controls, feature groups, leakage policy and delay lens
4	Analytical Approach	Regression, late-order classification, cost-sensitive policy and clustering
5	Innovation & Explainability	Differentiators, governance and operational value pathway
6	Recommendations & Risks	Three prioritised actions, limitations, conclusion and submission alignment

Prepared for: Hackathon evaluation

Project: Q-INTEL

1. Executive Summary

Quick-commerce operators make decisions under tight delivery-time expectations. A late order can create dissatisfaction, cancellation, refund and customer-retention risk. At the same time, store capacity and rider availability are costly resources. Managers need to understand delay patterns, identify risky orders early, evaluate geographic demand and support courier operations.

Q-INTEL addresses these needs through an end-to-end analytical workflow: data reliability checks; a transparent delivery-delay target; exploratory analysis; regression for delivery-time estimation; classification for late-order risk; geographic and courier clustering; and evidence-linked management actions.

Business objectives

Objective	Business question	Q-INTEL output
Delivery visibility	How long may an order take?	Predicted ETA in minutes
Late-order exposure	Which orders may breach the SLA?	Late-risk probability and decision flag
Demand geography	Where are orders concentrated?	Geographic clusters and zone profiles
Courier operations	Which courier groups may need support?	Courier segments and profiles
Management action	What should operations prioritise?	Three evidence-linked actions

Core analytical components

Supervised learning
Regression estimates continuous delivery duration; classification estimates the probability of crossing the defined service-level threshold.

Unsupervised learning
K-Means explores geographic demand groupings, while courier segmentation groups behavioural and performance profiles.

Decision support
Descriptive and predictive evidence is converted into operational priorities for managers.

Defining success

Success must be judged by more than accuracy. The solution should apply the following principles:

- Use **fair holdout evaluation** so reported performance reflects unseen data.
- **Avoid information leakage** by using only information available at prediction time.
- Select **metrics aligned with business costs**, not accuracy alone.
- **Explain assumptions** and separate measured outcomes from projected benefits.

Intended users
Operations managers, zone planners, delivery workforce managers and analytics teams. The prototype supports analysis and scenario discussion; it does not autonomously dispatch riders or open stores.

2. Data, Reliability & Feature Strategy

The challenge guide describes a public food-delivery dataset of roughly **45,000 orders** and **20 original columns**. Fields include store and customer coordinates, order date/time, weather, traffic, rider attributes, vehicle/order categories and observed delivery duration. The guide explicitly cautions that this is not a true grocery quick-commerce dataset; restaurant coordinates are used as store proxies.

Data reliability controls

Check	Treatment / control	Business reason
Missing values	Profile missingness and apply justified imputation within training pipelines where required.	Retain usable records without learning imputation values from holdout data.
Duplicates	Identify exact duplicate rows and document treatment.	Avoid double-counting repeated observations.
GPS validity	Flag invalid/sentinel/out-of-area coordinates; exclude invalid coordinates from spatial calculations.	Prevent false distances and misleading clusters.
Types / categories	Parse duration and timestamps; standardise labels; validate ages and ratings.	Make analytical fields consistent.
Outliers	Investigate implausible values and document retention, transformation or exclusion.	Avoid arbitrary deletion.

Feature groups

Family	Examples	How it helps
Geographic	Store/customer latitude and longitude; Haversine distance	Represents separation and spatial demand.
Temporal	Order hour, weekday, weekend, time period and peak-hour indicator where derived	Captures time-of-day and calendar patterns.
Environment	Weather, road-traffic density, festival indicator	Describes operating conditions associated with duration.
Order / vehicle	Order type, vehicle type and multiple deliveries when available at prediction time	Adds operational context; timing must be checked.
Courier profile	Rating, experience, average delivery time, late rate and deliveries/day for segmentation	Supports behavioural grouping and workforce planning.

Leakage policy
A model predicting at order placement must not use information only known after placement or delivery. Pickup timestamps and realised delivery outcomes are not valid pre-delivery predictors. Rider/dispatch fields belong in a separate prediction stage only if genuinely available then.

Delay lens
The revised notebook is described as using a **30-minute SLA** for the late-order label. Keep this definition consistent across training, evaluation, dashboard and report unless the challenge owner specifies otherwise.

3. Analytical Approach & Model Portfolio

Q-INTEL combines supervised and unsupervised learning because the business questions require different outputs. Regression estimates a continuous duration, classification estimates the probability of an SLA breach, and clustering discovers groups without predefined labels.

A. Delivery-time regression

Target: observed delivery duration in minutes. The current project direction compares a simple baseline with Ridge Regression, Random Forest Regressor and Gradient Boosting Regressor. The baseline indicates simple-reference performance; Ridge provides a regularised linear benchmark; tree ensembles can represent nonlinear relationships and feature interactions.

Model	Role	Selection evidence
Naive / Dummy baseline	Reference performance	Must be beaten to demonstrate model value.
Ridge Regression	Regularised linear model	MAE, RMSE and R^2 under the validation/test protocol.
Random Forest	Bagged decision-tree ensemble	Compare error and robustness against baseline.
Gradient Boosting	Sequential boosting ensemble	Check whether improved error justifies complexity.

Use **MAE** as the manager-friendly average error in minutes, **RMSE** to penalise large misses, and **R^2** to describe variance explained. Select on validation evidence, then report once on the untouched test set.

B. Late-order classification

Target: whether an order exceeds the 30-minute SLA. Evaluate with precision, recall, F1 and ROC-AUC; the precision-recall curve / PR-AUC is also useful when late orders are less frequent. Accuracy alone is insufficient because a model can appear accurate while missing late orders.

5 : 1
cost ratio assumption

Cost-sensitive decision policy

The revised approach frames a missed late order as **five cost units** and an unnecessary intervention as **one cost unit**. This 5:1 ratio is a business assumption to guide threshold selection. Tune the threshold on validation data and test sensitivity. It is not a measured financial cost unless supported by company data.

C. Unsupervised learning

Geographic clustering

K-Means groups order/store locations to explore candidate dark-store zones. Support cluster count with elbow and silhouette evidence. Compare zone demand, average delivery time and late rate with the network benchmark. A centroid is not automatically a real service boundary or approved store site.

Courier segmentation

Clustering on appropriate rider-level aggregates (rating, experience, average delivery time, late rate and deliveries/day) can reveal operational profiles. Segment names should be assigned only after inspecting actual cluster statistics.

4. Innovation, Explainability & Operational Value

Q-INTEL's differentiator is the combination of several decision views in one workflow rather than reliance on a single prediction. It links delivery-time estimation, late-risk screening, geographic analysis and courier segmentation to management decisions.

Key differentiators

Innovation	What is distinctive	Potential operational value
Dual prediction	Provides ETA and a separate late-risk probability.	Distinguishes long expected duration from elevated probability of SLA breach.
Cost-aware threshold	Uses a stated 5:1 missed-late vs unnecessary-intervention assumption.	Aligns alerts with business priorities rather than default 0.50.
Dark-store zone intelligence	Uses geographic clusters and zone-level operating measures.	Supports location-planning discussions and demand analysis.
Courier behavioural segments	Groups courier profiles using aggregated performance measures.	Can inform training, roster review and support plans.
ETA uncertainty objective	The notebook is described as including an 80% uncertainty-band objective; implementation and coverage need verification.	A range can communicate uncertainty better than a single point estimate.
Evidence-to-action workflow	Connects findings to three prioritised management actions.	Makes analytics actionable for stakeholders.

Explainability and governance

Use feature importance, permutation importance or SHAP **only where actually implemented and validated**. Importance indicates model reliance/association, not causation. Each recommendation should state the observed finding, supporting evidence, proposed action and pilot measurement.

1	2	3	4
Finding	Evidence	Action	Pilot measure

Operational value pathway

Analytical output	Possible management response	Pilot KPI
High late-risk score	Review order, store workload and available support; do not auto-dispatch without a validated policy.	Late-order recall, intervention precision, late rate
High-demand zone with weak service metrics	Investigate capacity, assortment, staffing and candidate-store feasibility.	Demand served, delivery radius, cost/order, SLA
Courier segment needing support	Offer coaching, review allocation or adjust shift support.	Segment late rate, workload balance, rating trend

Important
Q-INTEL is a decision-support prototype. It generates analytical signals and recommendations; it does not prove dispatch changes, courier interventions or store openings have been executed or produced savings.

5. Prioritised Recommendations, Risks & Conclusion

The following recommendations are framed as pilot actions. Final zone names, courier profiles and model findings must be inserted from the final executed notebook; do not use unverified cluster numbers or performance percentages.

Three prioritised management actions

Priority	Recommendation	Evidence required	Measure
1 — Protect service levels	Pilot a late-risk review queue. Review high-risk orders early and investigate store workload, distance or available courier capacity.	Validation-tested classifier; precision/recall at selected threshold; observed delay patterns.	Late rate, recall, false-alert workload and intervention cost.
2 — Investigate priority zones	Prioritise high-demand zones with delivery time or late rate worse than network benchmark for feasibility study — not automatic store opening.	K-Means evidence, zone volume, late rate, duration, GPS validity and cluster stability.	Demand served, delivery radius, cost/order, SLA, rent and capacity feasibility.
3 — Improve courier support	Use courier segments to identify groups that may benefit from coaching, balanced workload or peak-period roster review.	Rider-level aggregates, segment profiles, sample sizes and fair comparisons.	Late rate by segment, workload balance, rating trend and safety indicators.

Limitations and assumptions

- **Dataset proxy:** the source is food delivery, not grocery quick-commerce. Restaurant locations are store proxies; grocery picking, inventory availability, dark-store rent, basket complexity and fulfilment capacity are not fully represented.
- **Observational data:** associations between weather, traffic, rider attributes and delay do not establish causation.
- **Geographic limits:** coordinate errors and cluster scale can affect interpretation. Cluster centres are not service polygons or road-network routes.
- **Cost assumption:** the 5:1 ratio is an assumption and should be sensitivity-tested with actual business costs.
- **ETA uncertainty:** an 80% interval objective must be checked for empirical coverage and interval width on held-out data.
- **Deployment:** historical performance does not guarantee live performance; monitoring, retraining, privacy, fairness and integration remain necessary.

Conclusion

Q-INTEL provides a structured path from historical delivery records to predictive and operational intelligence. Its contribution is the integrated use of ETA regression, cost-aware late-risk classification, geographic demand clustering and courier segmentation, with attention to data quality, leakage prevention and business-facing recommendations. The next step is a controlled pilot using real quick-commerce data, validated routing and store-capacity information, and measured operational outcomes.

Submission alignment
Submit this report alongside the complete runnable notebook/source code, full model comparison table, Streamlit demo and README covering setup, dependencies, assumptions and execution. Keep final metric values and selected model names synchronised across all deliverables.